

Accent-dependent ASR Degradation After Speech Enhancement

Noise type: babble

SNR: 5 dB

Whisper model: openai/whisper-base.en

Enhancers: sepformer_dns4, metricgan_plus

Actual speakers per accent: {'American': 4, 'Arabic': 4, 'Hindi': 4, 'Korean': 4, 'Mandarin': 4, 'Spanish': 4, 'Vietnamese': 4}

Actual prompts per accent: {'American': 10, 'Arabic': 10, 'Hindi': 10, 'Korean': 10, 'Mandarin': 10, 'Spanish': 10, 'Vietnamese': 10}

Output directory: /content/accent_enhancement_audit_outputs

This report summarizes a matched-prompt smoke test for accent-dependent ASR degradation after controlled noise addition and speech enhancement. The key comparison is enhanced WER minus noisy WER. Positive values mean enhancement made Whisper worse than leaving the noisy audio unchanged. The noise-only section checks whether the controlled noise condition itself affects accent groups unevenly before enhancement.

Corpus WER/CER by Accent and Condition

accent	condition	n	corpus_wer	corpus_cer	mean_utt_wer	mean_utt_cer
American	metricgan_plus	40	0.365	0.263	0.375	0.277
American	noisy	40	1.574	1.422	1.860	1.896
American	raw	40	0.012	0.009	0.015	0.007
American	sepformer_dns4	40	0.185	0.130	0.183	0.126
Arabic	metricgan_plus	40	0.665	0.463	0.665	0.492
Arabic	noisy	40	0.488	0.349	0.452	0.313
Arabic	raw	40	0.159	0.080	0.142	0.070
Arabic	sepformer_dns4	40	0.447	0.287	0.430	0.276
Hindi	metricgan_plus	40	1.315	1.010	1.786	1.483
Hindi	noisy	40	0.506	0.386	0.512	0.386
Hindi	raw	40	0.062	0.045	0.050	0.036
Hindi	sepformer_dns4	40	0.424	0.286	0.440	0.305
Korean	metricgan_plus	40	0.647	0.447	0.641	0.444
Korean	noisy	40	1.662	1.392	1.213	0.909
Korean	raw	40	0.076	0.050	0.087	0.055
Korean	sepformer_dns4	40	0.306	0.204	0.320	0.214
Mandarin	metricgan_plus	40	0.606	0.419	0.599	0.422
Mandarin	noisy	40	0.503	0.414	0.477	0.373
Mandarin	raw	40	0.091	0.059	0.101	0.066
Mandarin	sepformer_dns4	40	1.394	1.215	2.143	2.088
Spanish	metricgan_plus	40	0.576	0.397	0.591	0.416
Spanish	noisy	40	0.444	0.315	0.461	0.335
Spanish	raw	40	0.106	0.069	0.124	0.077
Spanish	sepformer_dns4	40	1.053	0.919	1.206	1.106
Vietnamese	metricgan_plus	40	0.729	0.512	0.731	0.526
Vietnamese	noisy	40	0.632	0.476	0.633	0.469
Vietnamese	raw	40	0.206	0.119	0.208	0.119
Vietnamese	sepformer_dns4	40	0.624	0.421	0.634	0.427

Degradation Pivot: Enhanced vs Raw and Noisy

isy	raw	sepformer_dns4	metricgan_plus_delta_vs_raw	metricgan_plus_rel_vs_raw	metricgan_plus_delta_vs_noisy	metricgan_plus_rel_vs_noisy	sepformer_dns4_delta_vs_raw	sepformer_dns4_rel_vs_raw	sepformer_dns4_delta_vs_noisy
74	0.012	0.185	0.353	30.000	-1.209	-0.768	0.174	14.750	-1.388
88	0.159	0.447	0.506	3.185	0.176	0.361	0.288	1.815	-0.041
06	0.062	0.424	1.253	20.286	0.809	1.599	0.362	5.857	-0.082
62	0.076	0.306	0.571	7.462	-1.015	-0.611	0.229	3.000	-1.356
03	0.091	1.394	0.515	5.645	0.103	0.205	1.303	14.290	0.891
44	0.106	1.053	0.471	4.444	0.132	0.298	0.947	8.944	0.609

isy	raw	sepformer_dns4	metricgan_plus_delta_vs_raw	metricgan_plus_rel_vs_raw	metricgan_plus_delta_vs_noisy	metricgan_plus_rel_vs_noisy	sepformer_dns4_delta_vs_raw	sepformer_dns4_rel_vs_raw	sepformer_dns4_delta_vs_no
32	0.206	0.624	0.524	2.543	0.097	0.153	0.418	2.029	-0.009

Utterance-level Helped/Hurt Summary

accent	enhancer	n	mean_delta_utt_wer_vs_noisy	frac_hurt_vs_noisy	frac_helped_vs_noisy
American	sepformer_dns4	40	-1.677	0.125	0.350
Arabic	sepformer_dns4	40	-0.022	0.350	0.350
Hindi	sepformer_dns4	40	-0.072	0.250	0.575
Korean	sepformer_dns4	40	-0.893	0.075	0.475
Mandarin	sepformer_dns4	40	1.666	0.225	0.475
Spanish	sepformer_dns4	40	0.744	0.300	0.350
Vietnamese	sepformer_dns4	40	0.000	0.350	0.350
American	metricgan_plus	40	-1.485	0.500	0.250
Arabic	metricgan_plus	40	0.214	0.700	0.200
Hindi	metricgan_plus	40	1.273	0.650	0.200
Korean	metricgan_plus	40	-0.572	0.700	0.175
Mandarin	metricgan_plus	40	0.122	0.650	0.200
Spanish	metricgan_plus	40	0.130	0.625	0.175
Vietnamese	metricgan_plus	40	0.097	0.550	0.300

Noise-only Degradation Summary: Raw -> Noisy

accent	n	raw_mean_utt_wer	noisy_mean_utt_wer	mean_noise_delta_wer	median_noise_delta_wer	frac_noise_hurts
American	40	0.015	1.860	1.845	0.191	0.700
Korean	40	0.087	1.213	1.126	0.354	0.750
Hindi	40	0.050	0.512	0.463	0.462	0.825
Vietnamese	40	0.208	0.633	0.426	0.408	0.825
Mandarin	40	0.101	0.477	0.375	0.310	0.900
Spanish	40	0.124	0.461	0.337	0.364	0.825
Arabic	40	0.142	0.452	0.310	0.250	0.825

Noise Degradation vs Enhancement Effect

accent	mean_noise_delta_wer	mean_enh_delta_vs_noisy	enhancer
American	1.845	-1.677	sepformer_dns4
Arabic	0.310	-0.022	sepformer_dns4
Hindi	0.463	-0.072	sepformer_dns4
Korean	1.126	-0.893	sepformer_dns4
Mandarin	0.375	1.666	sepformer_dns4
Spanish	0.337	0.744	sepformer_dns4
Vietnamese	0.426	0.000	sepformer_dns4
American	1.845	-1.485	metricgan_plus
Arabic	0.310	0.214	metricgan_plus
Hindi	0.463	1.273	metricgan_plus
Korean	1.126	-0.572	metricgan_plus
Mandarin	0.375	0.122	metricgan_plus
Spanish	0.337	0.130	metricgan_plus
Vietnamese	0.426	0.097	metricgan_plus

Worst Noise-only Failures

accent	speaker	utt_id	reference	metricgan_plus	noisy	raw	sepformer_dns4	noise_delta_wer	noise_rel_delta_wer	noise_hurts
American	clb	anyway no one saw her like that	Anyway, no one saw her like that.	0.571	63.429	0.000	0.143	63.429	63428571428571.430	True
Korean	YKWK	hardly were our plans made...	hardly were our plans made...	0.692	34.154	0.077	0.615	34.077	443.000	True
Hindi	SVBI	you must sleep he urged	you must sleep he urged	44.400	1.200	0.000	0.400	1.200	1200000000000.000	True
Arabic	ZHAA	hardly were our plans made...	hardly were our plans made...	0.846	1.462	0.308	0.846	1.154	3.750	True
Mandarin	BWC	i had faith in them	i had faith in them	0.400	1.000	0.000	1.000	1.000	1000000000000.000	True
Hindi	RRBI	you must sleep he urged	you must sleep he urged	1.000	1.000	0.000	1.000	1.000	1000000000000.000	True
Mandarin	NCC	hardly were our plans made...	hardly were our plans made...	0.538	1.000	0.000	0.385	1.000	1000000000000.000	True
Spanish	ERMS	there has been a change she...	there has been a change she...	1.000	1.125	0.125	0.500	1.000	8.000	True

accent	speaker	utt_id	reference	metricgan_plus	noisy	raw	sepformer_dns4	noise_delta_wer	noise_rel_delta_wer	noise_hurts
Vietnamese	TLV	anyway no one saw her like that	anyway no one saw her like that	1.143	1.000	0.000	0.429	1.000	100000000000.000	True
Vietnamese	THV	he obeyed the pressure of her hand	he obeyed the pressure of her hand	1.571	1.000	0.000	0.714	1.000	100000000000.000	True
Hindi	TNI	there has been a change she...	there has been a change she...	0.375	1.000	0.000	0.250	1.000	100000000000.000	True
American	clb	there has been a change she...	There has been a change, she...	0.500	1.000	0.000	0.500	1.000	100000000000.000	True
Mandarin	BWC	hardly were our plans made...	hardly were our plans made...	0.385	1.000	0.077	0.154	0.923	12.000	True
Mandarin	LXC	he had no excuse for the...	he had no excuse for the...	0.750	0.917	0.000	0.167	0.917	91666666666.667	True
American	clb	he had no excuse for the...	He had no excuse for the...	0.333	0.917	0.000	0.000	0.917	91666666666.667	True
American	bdl	he had no excuse for the...	He had no excuse for the...	0.250	0.917	0.000	0.917	0.917	91666666666.667	True
Hindi	TNI	how could he explain his...	how could he explain his...	0.889	0.889	0.000	0.778	0.889	88888888888.889	True
Mandarin	BWC	there has been a change she...	there has been a change she...	0.625	1.000	0.125	1.000	0.875	7.000	True
Mandarin	TXHC	there has been a change she...	there has been a change she...	1.000	0.875	0.000	0.750	0.875	87500000000.000	True
Korean	YKWK	he obeyed the pressure of her hand	he obeyed the pressure of her hand	0.857	1.000	0.143	0.429	0.857	6.000	True

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Mean Signal Diagnostics by Accent and Condition

accent	condition	duration_sec	rms	peak	clipped_frac
American	metricgan_plus	2.746	0.137	1.000	0.000
American	noisy	2.746	0.122	0.691	0.000
American	raw	2.746	0.106	0.613	0.000
American	sepformer_dns4	2.746	0.167	0.990	0.000
Arabic	metricgan_plus	3.044	0.135	1.000	0.000
Arabic	noisy	3.044	0.118	0.680	0.000
Arabic	raw	3.044	0.104	0.604	0.000
Arabic	sepformer_dns4	3.044	0.164	0.990	0.000
Hindi	metricgan_plus	2.760	0.131	1.000	0.000
Hindi	noisy	2.760	0.114	0.642	0.000
Hindi	raw	2.760	0.099	0.580	0.000
Hindi	sepformer_dns4	2.760	0.172	0.990	0.000
Korean	metricgan_plus	3.022	0.140	1.000	0.000
Korean	noisy	3.022	0.121	0.777	0.000
Korean	raw	3.022	0.111	0.699	0.000
Korean	sepformer_dns4	3.022	0.149	0.990	0.000
Mandarin	metricgan_plus	3.343	0.138	1.000	0.000
Mandarin	noisy	3.343	0.112	0.645	0.000
Mandarin	raw	3.343	0.100	0.595	0.000
Mandarin	sepformer_dns4	3.343	0.169	0.990	0.000
Spanish	metricgan_plus	3.349	0.121	1.000	0.000
Spanish	noisy	3.349	0.099	0.626	0.000
Spanish	raw	3.349	0.086	0.570	0.000
Spanish	sepformer_dns4	3.349	0.150	0.990	0.000
Vietnamese	metricgan_plus	3.212	0.137	1.000	0.000
Vietnamese	noisy	3.212	0.109	0.578	0.000
Vietnamese	raw	3.212	0.095	0.504	0.000
Vietnamese	sepformer_dns4	3.212	0.185	0.990	0.000

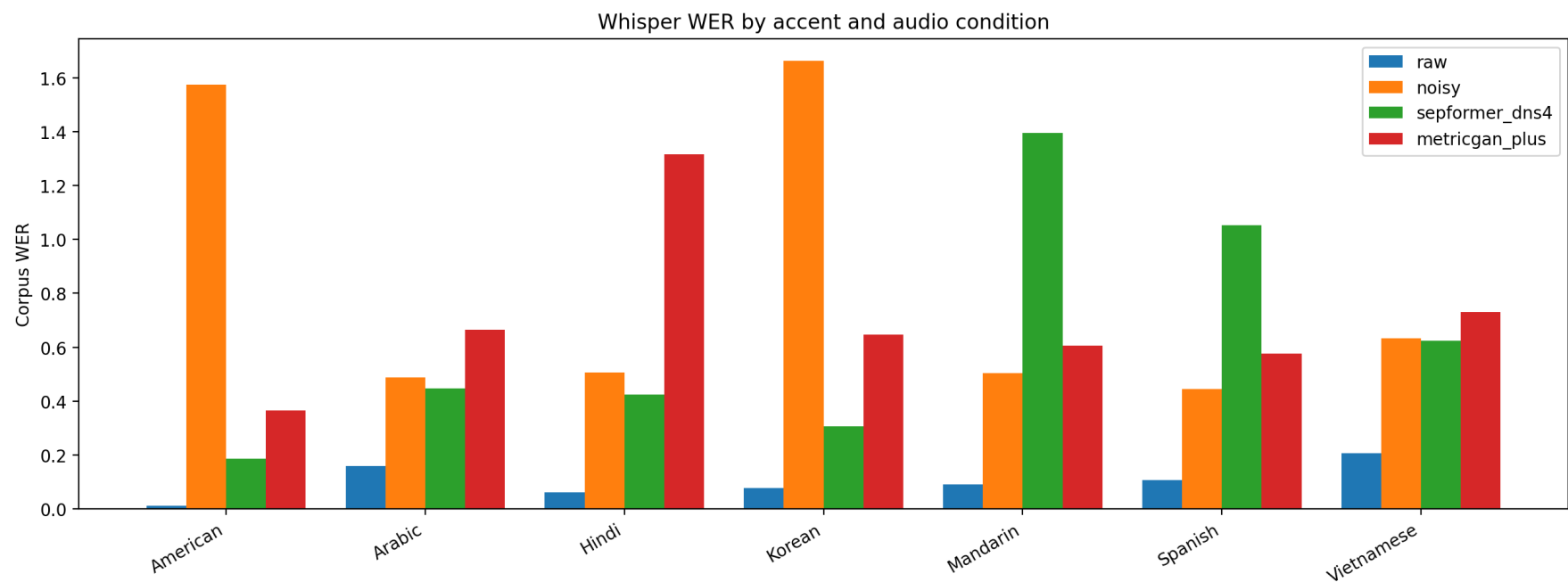
Signal Diagnostics for Worst Failure Cases

accent	speaker	utt_id	condition	duration_sec	rms	peak	clipped_frac	path	failure_type
American	clb	anyway no one saw her like that	raw	2.375	0.149	0.650	0.000	...	noise_only
American	clb	anyway no one saw her like that	noisy	2.375	0.172	0.874	0.000	...	noise_only
American	clb	anyway no one saw her like that	sepformer_dns4	2.375	0.188	0.990	0.000	...	noise_only
American	clb	anyway no one saw her like that	metricgan_plus	2.375	0.129	1.000	0.000	...	noise_only
Korean	YKWK	hardly were our plans made...	raw	5.680	0.124	0.857	0.000	...	noise_only
Korean	YKWK	hardly were our plans made...	noisy	5.680	0.123	0.990	0.000	...	noise_only

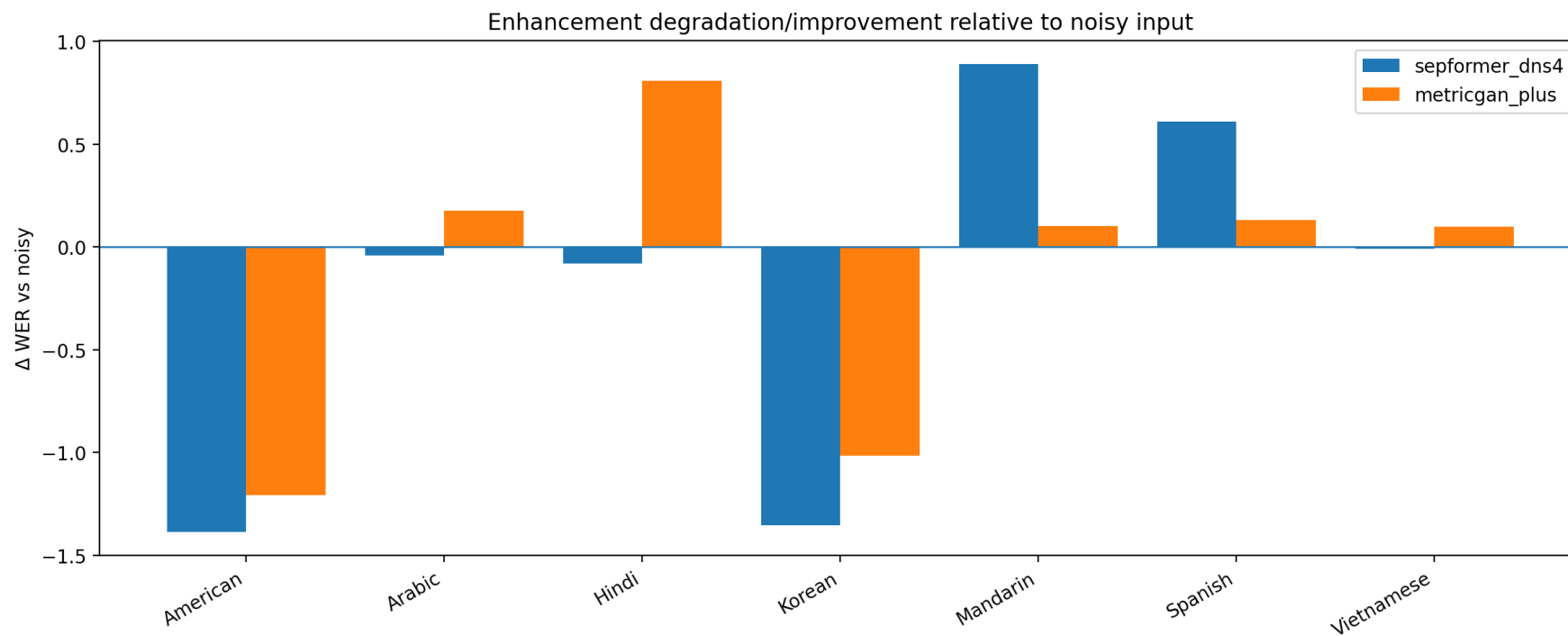
accent	speaker	utt_id	condition	duration_sec	rms	peak	clipped_frac	path	failure_type
Korean	YKWK	hardly were our plans made...	sepformer_dns4	5.680	0.120	0.990	0.000	...	noise_only
Korean	YKWK	hardly were our plans made...	metricgan_plus	5.680	0.128	1.000	0.000	...	noise_only
Hindi	SVBI	you must sleep he urged	raw	1.517	0.111	0.496	0.000	...	noise_only
Hindi	SVBI	you must sleep he urged	noisy	1.517	0.128	0.571	0.000	...	noise_only
Hindi	SVBI	you must sleep he urged	sepformer_dns4	1.517	0.214	0.990	0.000	...	noise_only
Hindi	SVBI	you must sleep he urged	metricgan_plus	1.517	0.119	1.000	0.000	...	noise_only
Arabic	ZHAA	hardly were our plans made...	raw	5.234	0.072	0.619	0.000	...	noise_only
Arabic	ZHAA	hardly were our plans made...	noisy	5.234	0.083	0.682	0.000	...	noise_only
Arabic	ZHAA	hardly were our plans made...	sepformer_dns4	5.234	0.109	0.990	0.000	...	noise_only
Arabic	ZHAA	hardly were our plans made...	metricgan_plus	5.234	0.079	1.000	0.000	...	noise_only
Mandarin	BWC	i had faith in them	raw	2.145	0.081	0.489	0.000	...	noise_only
Mandarin	BWC	i had faith in them	noisy	2.145	0.094	0.570	0.000	...	noise_only
Mandarin	BWC	i had faith in them	sepformer_dns4	2.145	0.160	0.990	0.000	...	noise_only
Mandarin	BWC	i had faith in them	metricgan_plus	2.145	0.131	1.000	0.000	...	noise_only
Hindi	RRBI	you must sleep he urged	raw	1.720	0.099	0.457	0.000	...	noise_only
Hindi	RRBI	you must sleep he urged	noisy	1.720	0.116	0.487	0.000	...	noise_only
Hindi	RRBI	you must sleep he urged	sepformer_dns4	1.720	0.249	0.990	0.000	...	noise_only
Hindi	RRBI	you must sleep he urged	metricgan_plus	1.720	0.130	1.000	0.000	...	noise_only
Mandarin	NCC	hardly were our plans made...	raw	6.267	0.098	0.575	0.000	...	noise_only
Mandarin	NCC	hardly were our plans made...	noisy	6.267	0.112	0.629	0.000	...	noise_only
Mandarin	NCC	hardly were our plans made...	sepformer_dns4	6.267	0.166	0.990	0.000	...	noise_only
Mandarin	NCC	hardly were our plans made...	metricgan_plus	6.267	0.137	1.000	0.000	...	noise_only
Spanish	ERMS	there has been a change she...	raw	3.525	0.080	0.463	0.000	...	noise_only
Spanish	ERMS	there has been a change she...	noisy	3.525	0.093	0.504	0.000	...	noise_only
Spanish	ERMS	there has been a change she...	sepformer_dns4	3.525	0.162	0.990	0.000	...	noise_only
Spanish	ERMS	there has been a change she...	metricgan_plus	3.525	0.112	1.000	0.000	...	noise_only
Vietnamese	TLV	anyway no one saw her like that	raw	2.340	0.094	0.546	0.000	...	noise_only
Vietnamese	TLV	anyway no one saw her like that	noisy	2.340	0.108	0.609	0.000	...	noise_only
Vietnamese	TLV	anyway no one saw her like that	sepformer_dns4	2.340	0.191	0.990	0.000	...	noise_only
Vietnamese	TLV	anyway no one saw her like that	metricgan_plus	2.340	0.130	1.000	0.000	...	noise_only
Vietnamese	THV	he obeyed the pressure of her hand	raw	2.390	0.119	0.707	0.000	...	noise_only
Vietnamese	THV	he obeyed the pressure of her hand	noisy	2.390	0.136	0.823	0.000	...	noise_only
Vietnamese	THV	he obeyed the pressure of her hand	sepformer_dns4	2.390	0.163	0.990	0.000	...	noise_only
Vietnamese	THV	he obeyed the pressure of her hand	metricgan_plus	2.390	0.135	1.000	0.000	...	noise_only

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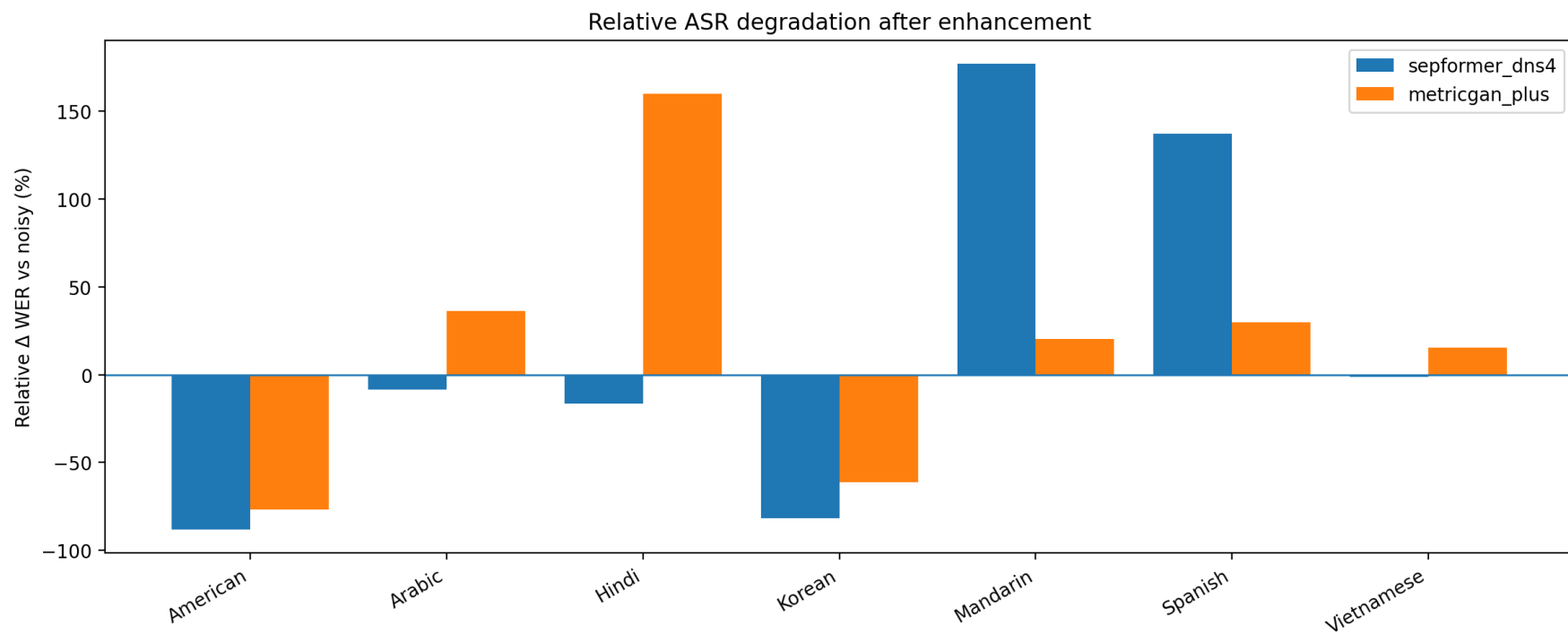
WER by Accent and Condition



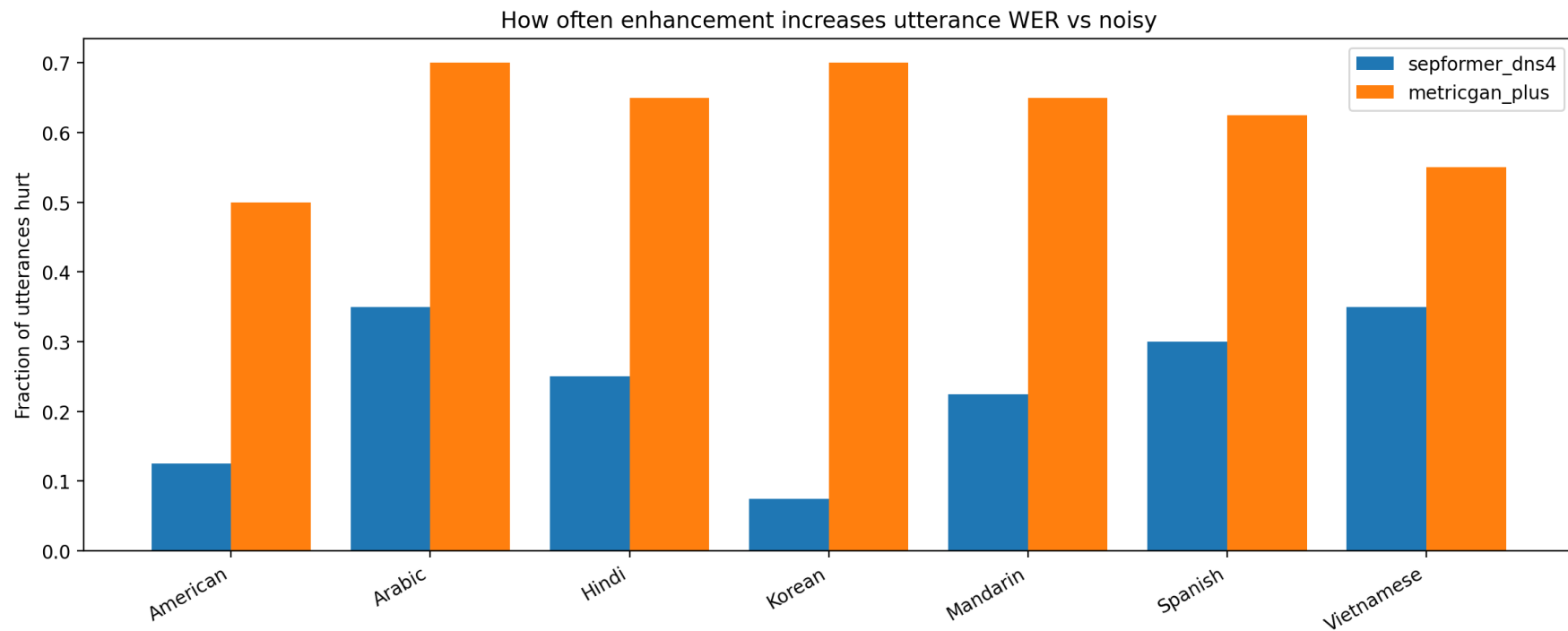
Delta WER vs Noisy



Relative Delta WER vs Noisy

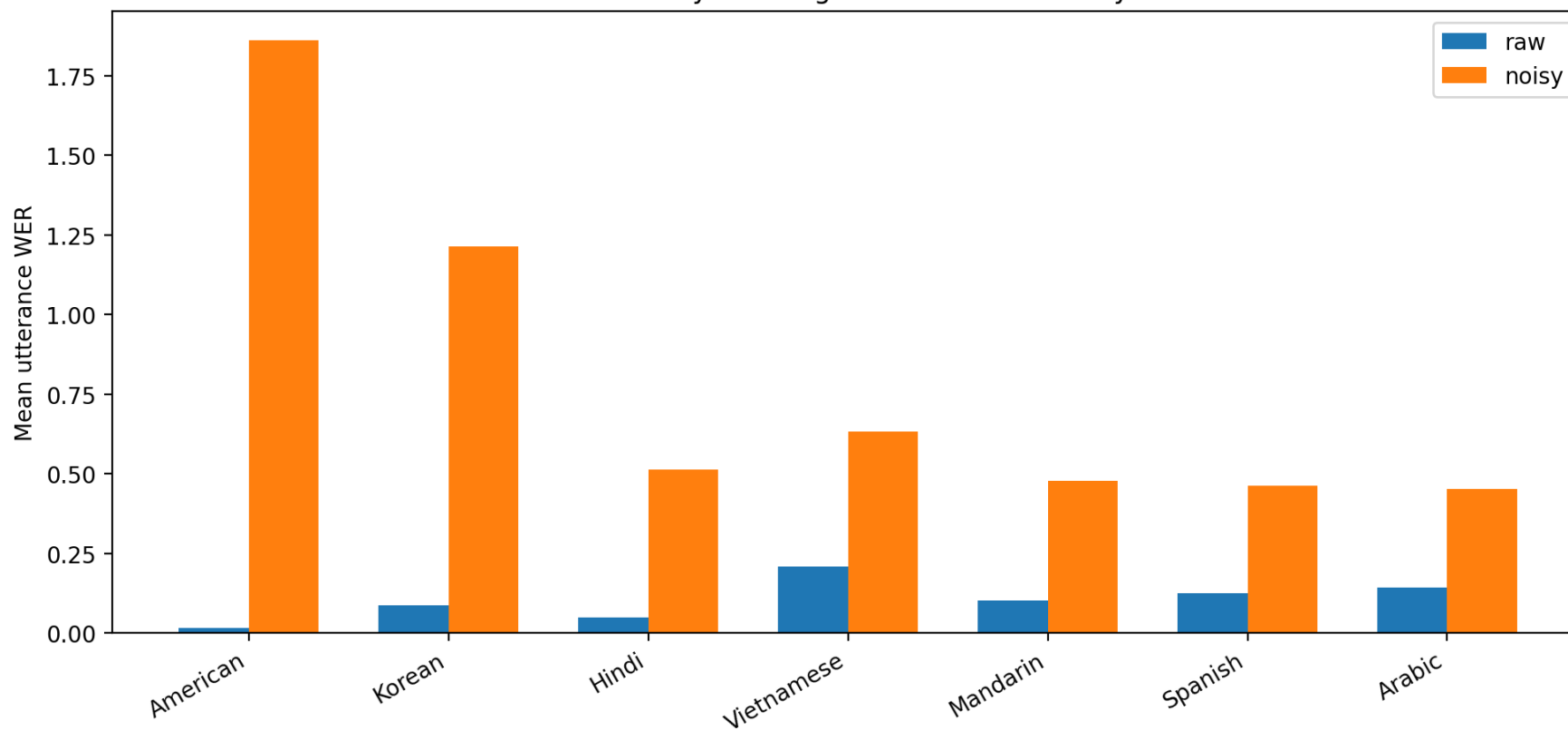


Fraction Hurt vs Noisy



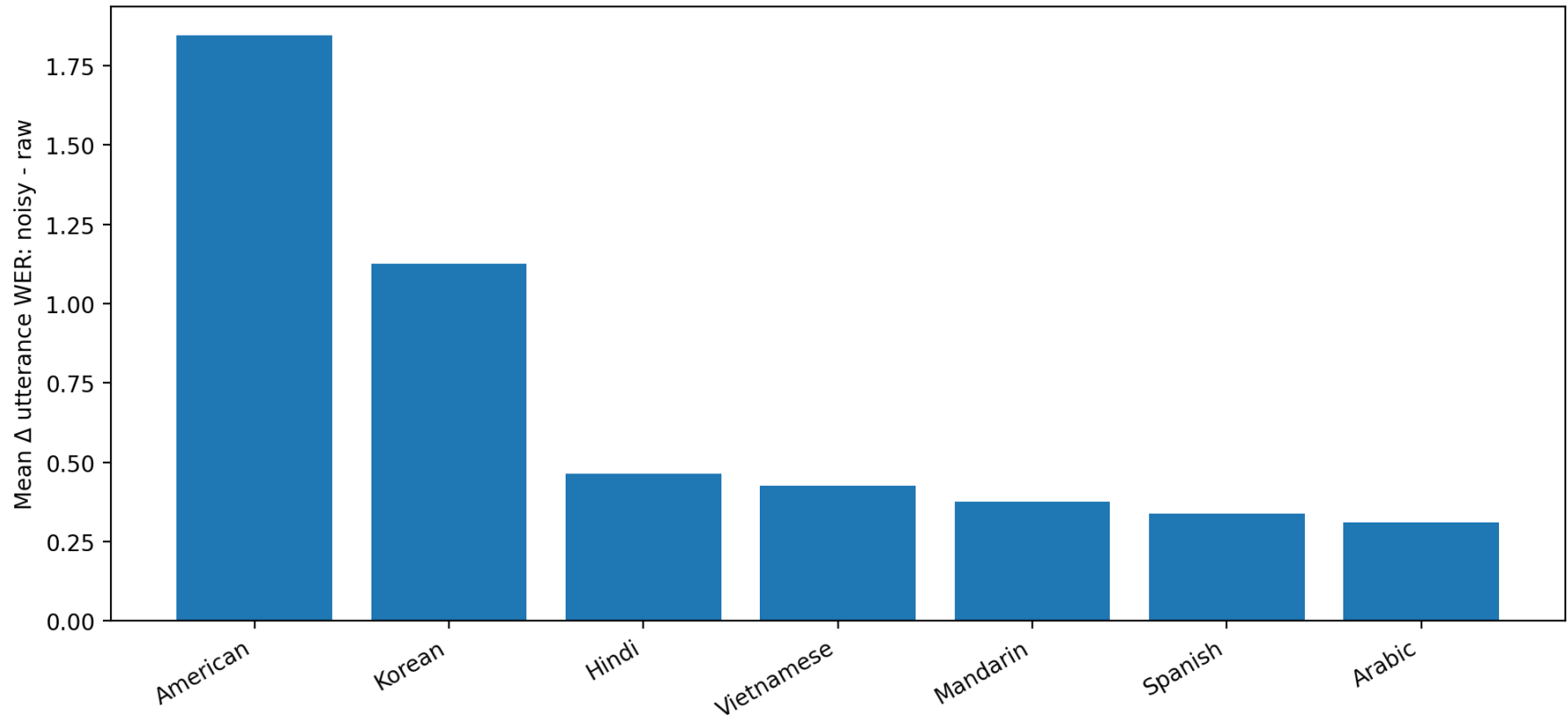
Noise Raw vs Noisy by Accent

Noise-only ASR degradation: raw vs noisy

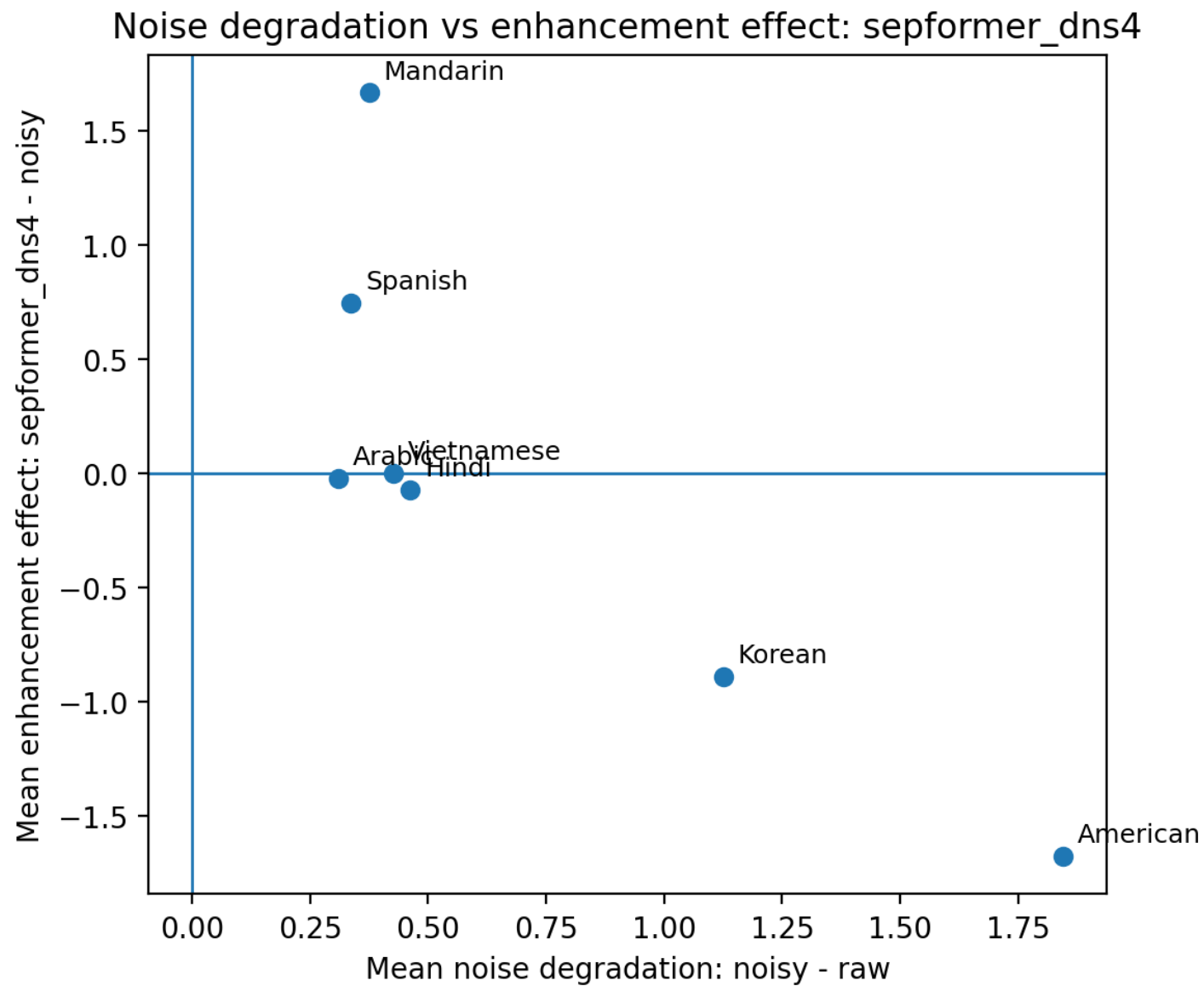


Noise Delta WER by Accent

How much does controlled noise degrade Whisper ASR?



Noise Degradation vs Enhancement Effect: sepformer_dns4



Noise Degradation vs Enhancement Effect: metricgan_plus

Noise degradation vs enhancement effect: metricgan_plus

